High-Precision Structural Mesh Reconstruction: A Hybrid Neural-Geometric Approach

NIO University Data Challenge Technical Report

Abstract: This report outlines the methodology, tools, and AI models used to build the NIO University Data Challenge pipeline. The objective is to automatically process 2D JPEG images and reconstruct them into the `.nas` format. The workflow combines ROI-focused preprocessing, HAWPv3-based wireframe parsing, geometric refinement, coordinate calibration, and final element-validity filtering before `.nas` serialization.

1. Methodology

1. **Preprocessing & Axis Extraction:** We employ color-space masking and connected component labeling, via morphological closing to isolate the structural-grid ROI from magenta axis artifacts. A custom `roi_sharp` lossless enhancement is then applied; empirical testing shows that this sharpness balance reduces interference for downstream networks.
2. **Optimized Wireframe Parsing:** We conducted tests using different methods and models, and ultimately found that the best performance was achieved in this mode. We run the HAWPv3 network with a dataset-tuned configuration. To maximize edge-detection recall, we process two fixed input variants (`roi_sharp` and `roi_sharp_context`) at asymmetric high resolution (1792×1024), together with 0.05 confidence.
3. **Geometric Graph Refinement:** To bridge the gap between probabilistic neural predictions and deterministic finite element meshes, we engineered a rigorous topological cleanup pipeline. This includes spatial clustering for endpoint snapping, optical-marker-aware vertex calibration, and confidence-based pruning of duplicate parallel edges. Furthermore, we enforce strict structural integrity via cross-product intersection splitting, Ramer-Douglas-Peucker collinear decimation, and recursive folding of dangling ears and degenerate short edges to guarantee closed-loop topologies.
4. **Mesh Recovery & Optimization:** Face candidates are extracted through planar cycle detection. To resolve overlaps, we formulate face selection as a Boolean optimization problem solved by Google OR-Tools CP-SAT, with strict mutual-exclusion constraints. Elements are assembled into compliant `CQUAD4`/`CTRIA3` polygons using polar-angle node sorting. Finally, centroid RGB sampling acts as a dark-region filter, automatically removing phantom background faces (e.g. Image 87).
5. **Stable Coordinate Mapping:** Pixel coordinates are mapped to physical space through a robust correction chain. The chain combines per-image magenta-axis mapping with offline ground-truth calibration using affine and similarity transformation matrices. A bounding-box penalty and L2-norm heuristic dynamically routes between sub-pixel matrix correction and fallback axis-template stabilization.
6. **NAS Format Generation:** The final validated mesh topology is serialized into strict `.nas` syntax and exported.

2. AI & LLM Usage Disclosure

During planning, research, development, and debugging, we used AI and large language models to accelerate implementation:

- **ChatGpt 5.4:** Served as an interactive development assistant to aid in rapid prototyping, provide contextual code suggestions, and troubleshoot syntax.
- **OpenAI-Prism:** Using Prism under OPENAI for LaTeX code correction and multi person collaboration.
- **Google Gemini 3.1 Pro:** Applied for copyediting, refining grammar, and enhancing the overall clarity and fluency of this report.

All AI-generated snippets were reviewed, manually tested, and integrated by human developers to satisfy the geometric constraints of the target `.nas` format.